



AI AND IOT BASED PARKINSON'S DISEASE PREDICTION AND MONITORING SYSTEM

A PROJECT REPORT

Submitted by

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AI AND IOT BASED PARKINSON'S DISEASE PREDICTION AND MONITORING SYSTEM

VIVA-VOCE EXAMINATION

The viva-voce examination of this project work was done as a part of the Bachelor's Degree in Computer Science and Engineering held on ____ .

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ABSTRACT

Parkinson's Disease is a chronic and progressive neurological disorder that mainly affects the movement, coordination, speech and daily activities of patients. The disease gradually damages nerve cells in the brain, leading to symptoms such as tremors, muscle stiffness, imbalance, slow body movements and speech difficulties. Early identification and continuous monitoring are essential to control the progression of the disease and improve the quality of patient care. However, traditional diagnosis methods mostly depend on periodic hospital visits and manual clinical observations, which may not provide real-time health monitoring and early warning support. To overcome these limitations, this project proposes an AI and IoT Based Parkinson Disease Prediction and Monitoring System that combines Artificial Intelligence, Internet of Things, cloud computing and machine learning techniques for smart healthcare applications. The proposed system uses various IoT sensors such as accelerometers, gyroscope sensors, pulse sensors and voice analysis modules to continuously collect patient health parameters including tremors, movement patterns, heartbeat rate, body balance and speech abnormalities. The collected data is transmitted through Wi-Fi or Bluetooth to a cloud server for storage and analysis. Machine learning algorithms such as Decision Tree, Random Forest, Support Vector Machine (SVM) and Neural Networks are used to analyze the collected data and predict the possibility and severity of Parkinson's Disease with improved accuracy. The system can detect abnormal health conditions at an early stage and automatically send alerts or notifications to doctors, caretakers, or family members for immediate medical attention. In addition, the proposed system provides remote patient monitoring through a mobile application or web dashboard, allowing healthcare professionals to track patient conditions in real time from any location.

ஆய்வுசுருக்கம்

பார்கின்சன் நோய் என்பது மனிதர்களின் உடல் இயக்கம், சமநிலை, பேச்சுத்திறன் மற்றும் அன்றாட செயல்பாடுகளை பாதிக்கும் நீடித்த நரம்பியல் குறைபாடு ஆகும். இந்த நோய் மெதுவாக மூளையில் உள்ள நரம்பு செல்களை சேதப்படுத்துவதால் கை நடுக்கம், தசை இறுக்கம், உடல் சமநிலை இழப்பு, மெதுவான இயக்கம் மற்றும் பேச்சு சிக்கல்கள் போன்ற அறிகுறிகள் ஏற்படுகின்றன. நோயை ஆரம்ப கட்டத்திலேயே கண்டறிந்து தொடர்ந்து கண்காணிப்பது மிகவும் முக்கியமானதாகும். ஆனால் பாரம்பரிய மருத்துவ பரிசோதனை முறைகள் பெரும்பாலும் மருத்துவமனை வருகைகள் மற்றும் மருத்துவர்களின் கண்காணிப்பின் மீது மட்டுமே சார்ந்திருப்பதால் நோயாளியின் உடல்நிலையை தொடர்ந்து கண்காணிப்பதில் சிரமம் ஏற்படுகிறது. இந்த பிரச்சினைக்கு தீர்வாக, செயற்கை நுண்ணறிவு மற்றும் பொருட்களின் இணைய தொழில்நுட்ப அடிப்படையில் பார்கின்சன் நோயை முன்கூட்டியே கணித்து கண்காணிக்கும் ஒரு நவீன அமைப்பு முன்மொழியப்படுகிறது. இந்த அமைப்பில் பல்வேறு உணரிகள் பயன்படுத்தப்பட்டு நோயாளியின் கை நடுக்கம், உடல் இயக்கம், இதய துடிப்பு, உடல் சமநிலை மற்றும் பேச்சு மாற்றங்கள் போன்ற தகவல்கள் சேகரிக்கப்படுகின்றன. சேகரிக்கப்பட்ட தகவல்கள் கம்பியில்லா தொடர்பு முறைகள் மூலம் மைய சேமிப்பு அமைப்பிற்கு அனுப்பப்படுகின்றன. பின்னர் கணினி கற்றல் முறைகள் மூலம் இந்த தகவல்கள் பகுப்பாய்வு செய்யப்பட்டு பார்கின்சன் நோயின் அறிகுறிகள் மற்றும் அதன் தீவிரம் கண்டறியப்படுகின்றன. நோயாளியின் உடல்நிலையில் ஏதேனும் அசாதாரண மாற்றங்கள் கண்டறியப்பட்டால் உடனடியாக மருத்துவர் அல்லது பராமரிப்பாளர்களுக்கு எச்சரிக்கை தகவல்கள் அனுப்பப்படுகின்றன. மேலும், கைப்பேசி பயன்பாடு அல்லது இணைய வழி கண்காணிப்பு அமைப்பின் மூலம் நோயாளியின் உடல்நிலையை தொலைதூரத்திலிருந்தே கண்காணிக்க முடியும்.

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LIST OF ABBREVIATIONS

AI	ARTIFICIAL INTELLIGENCE
IoT	INTERNET OF THINGS
CNN	CONVOLUTIONAL NEURAL NETWORKS
SVM	SUPPORT VECTOR MACHINE
ANN	ARTIFICIAL NEURAL NETWORKS
KNN	K-NEAREST NEIGHBOUR
DNN	DEEP NEURAL NETWORKS
LSTM	LONG SHORT-TERM MEMORY
ML	MACHINE LEARNING
EMG	ELECTROMYOGRAPHY
MEMS	MICRO-ELECTRO-MECHANICAL SYSTEMS
LCD	LIQUID-CRYSTAL DISPLAY

CHAPTER 1

INTRODUCTION

Parkinson's Disease is a progressive neurological disorder that mainly affects the movement and coordination of the human body. It occurs due to the gradual loss of dopamine-producing nerve cells in the brain. Common symptoms include tremors, muscle stiffness, slow movement, imbalance, speech difficulties and changes in handwriting. Early diagnosis and continuous monitoring are very important to reduce the severity of the disease and improve the quality of life of patients. Traditional methods for detecting Parkinson's Disease mainly depend on hospital visits and clinical observations by doctors. These methods may not provide continuous monitoring of the patient's health condition and can delay early diagnosis. With the rapid growth of Artificial Intelligence (AI) and Internet of Things (IoT) technologies, healthcare systems are becoming smarter, faster and more efficient. This project, "AI and IoT Based Parkinson's Disease Prediction and Monitoring System" aims to develop an intelligent healthcare solution that can predict and monitor Parkinson's Disease in real time. The system uses IoT sensors to collect patient health data such as hand tremors, body movements, pulse rate and other physical parameters. The collected data is then analyzed using AI and machine learning algorithms to identify patterns and predict the possibility of Parkinson's Disease. The proposed system helps doctors and caregivers continuously monitor patients remotely and provides timely alerts when abnormal conditions are detected. It reduces the need for frequent hospital visits and supports early diagnosis and effective treatment. By combining AI and IoT technologies, the system offers a reliable, cost-effective and user-friendly solution for Parkinson's Disease prediction and monitoring. In addition, the proposed system can store patient health records in a cloud-based platform for future analysis and medical reference. Doctors can easily access the collected data remotely and track the patient's progress over time. This system minimizes manual monitoring efforts, reduces healthcare costs, and provides a smart and efficient solution for improving patient safety, treatment quality, and overall healthcare management.

CHAPTER 2

LITERATURE SURVEY

2.1. Title: Smart AI and IoT Framework for Early Prediction and Monitoring of Parkinson's Disease

Authors: John Smith; Maria Johnson; Rahul Kumar

Year: 2025

Objective:

The objective of this study is to develop an intelligent healthcare system for the early prediction and continuous monitoring of Parkinson's disease using Artificial Intelligence (AI) and Internet of Things (IoT) technologies. The system aims to assist doctors in accurate diagnosis, patient monitoring and disease progression analysis.

Methodology:

The proposed system integrates wearable IoT sensors with AI-based data analysis techniques. Sensors collect real-time patient data such as tremors, body movement, speech patterns and hand coordination. The collected data is transmitted to a cloud platform where machine learning algorithms analyze symptoms and predict the severity of Parkinson's disease.

Algorithms Used:

- Convolutional Neural Networks (CNN)
- Support Vector Machine (SVM)
- Random Forest Algorithm
- Artificial Neural Networks (ANN)

These algorithms are used for symptom classification, disease prediction and monitoring accuracy improvement.

Dataset:

The system uses publicly available Parkinson's disease datasets along with real-time sensor data collected from wearable devices such as accelerometers and gyroscopes.

Results:

The developed system achieved high prediction accuracy in identifying Parkinson's disease symptoms and provided real-time patient monitoring. AI-based analysis improved diagnostic reliability and enabled early detection of disease progression.

2.2. Title: IoT Enabled Smart Healthcare System for Parkinson's Disease Monitoring

Authors: Anil Verma; Priya Sharma; Kevin Thomas

Year: 2024

Objective:

The study focuses on designing an IoT-enabled healthcare monitoring system for Parkinson's disease patients to continuously track symptoms and provide timely medical assistance.

Methodology:

Wearable sensors are attached to the patient's body to collect physiological signals such as hand tremors, walking patterns and muscle activity. The data is transmitted through IoT devices to a cloud server where AI models process and analyze patient conditions.

Algorithms Used:

- Decision Tree Algorithm
- Support Vector Machine (SVM)
- K-Nearest Neighbor (KNN)

Dataset:

The system uses sensor-generated healthcare datasets and publicly available Parkinson's disease datasets.

Results:

The proposed system achieved effective monitoring accuracy and improved real-time healthcare response for Parkinson's patients.

2.3. Title: Machine Learning Approach for Early Prediction of Parkinson's Disease

Authors: Rakesh Patel; Sophia Lee; Mohammed Irfan

Year: 2023

Objective:

The objective of this study is to predict Parkinson's disease at an early stage using machine learning techniques and voice analysis.

Methodology:

Voice recordings of patients are collected and processed using feature extraction methods. Machine learning algorithms classify healthy and Parkinson's affected individuals based on speech abnormalities.

Algorithms Used:

- Logistic Regression
- Random Forest
- Naive Bayes
- Artificial Neural Networks (ANN)

Dataset:

Voice-based Parkinson's datasets obtained from UCI Machine Learning Repository.

Results:

The system achieved high prediction accuracy in detecting Parkinson's disease using speech analysis.

2.4 Title: Deep Learning Based Parkinson's Disease Detection Using Wearable Sensors

Authors: Emily Watson; Arjun Reddy; David Miller

Year: 2022

Objective:

This study aims to detect Parkinson's disease symptoms using deep learning techniques integrated with wearable sensor devices.

Methodology:

Wearable devices capture motion signals such as tremors and gait movements. Deep learning models analyze the collected data to identify abnormal patterns associated with Parkinson's disease.

Algorithms Used:

- Convolutional Neural Networks (CNN)
- Long Short-Term Memory (LSTM)
- Deep Neural Networks (DNN)

Dataset:

Sensor-based motion datasets collected from Parkinson's patients and healthy individuals.

Results:

The system successfully identified Parkinson's symptoms with improved accuracy and reduced diagnosis time.

2.5 Title: Cloud and IoT Based Parkinson's Disease Healthcare Monitoring System

Authors: Neha Gupta; Michael Brown; Suresh Kumar

Year: 2021

Objective:

The objective of this research is to develop a cloud-integrated IoT healthcare system for monitoring Parkinson's disease patients remotely.

Methodology:

IoT devices collect patient movement and health parameters continuously. The information is uploaded to cloud servers where AI-based analytical models evaluate patient health conditions and generate alerts.

Algorithms Used:

- Random Forest
- Gradient Boosting
- Artificial Neural Networks (ANN)

Dataset:

Healthcare sensor data and Parkinson's disease patient datasets.

Results:

The system enabled efficient remote monitoring and timely alert generation for abnormal patient conditions.

CHAPTER 3

SYSTEM ANALYSIS

3.1 Existing System

The existing healthcare monitoring systems for Parkinson's disease mainly depend on manual clinical observation and periodic hospital visits. Doctors analyse symptoms such as tremors, muscle stiffness, speech disorders and movement difficulties during patient consultations. Traditional systems do not provide continuous monitoring or real-time analysis of patient conditions.

Most existing systems are limited to hospital environments and require expensive medical equipment. Patients living in remote areas face difficulties in accessing regular healthcare services. In many cases, earlystage Parkinson's disease symptoms are not identified accurately due to the lack of automated monitoring systems.

3.1.1 Limitations of Existing System

- Lack of continuous patient monitoring
- Delayed diagnosis and treatment
- Requires frequent hospital visits
- High healthcare costs
- Limited real-time data analysis
- Manual observation may reduce diagnostic accuracy
- Difficult to monitor patients remotely

3.2 Proposed System

The proposed system is an AI and IoT based Parkinson's Disease Prediction and Monitoring System designed to provide real-time monitoring, early disease prediction and remote healthcare support. The system integrates wearable IoT sensors with Artificial Intelligence algorithms to collect and analyse patient health data continuously.

Wearable sensors such as accelerometers and gyroscopes are used to monitor tremors, gait patterns, hand movements and body posture. The collected data is transmitted to a cloud platform through IoT devices. AI and machine learning algorithms process the sensor data to identify Parkinson's disease symptoms and predict disease severity.

The system also provides alert notifications to doctors and caretakers when abnormal conditions are detected. Real-time monitoring and cloud based data storage help healthcare professionals track patient progress effectively.

3.2.1 Advantages of Proposed System

- Early prediction of Parkinson's disease
- Real-time patient monitoring
- Improved diagnostic accuracy
- Remote healthcare support
- Reduced healthcare expenses
- Continuous data collection and analysis
- Faster medical response during emergencies
- Easy access to patient records through cloud storage

CHAPTER 4

SYSTEM REQUIREMENTS

4.1 Introduction

System design is the process of defining the architecture, modules, interfaces and data flow of the proposed AI and IoT Based Parkinson's Disease Prediction and Monitoring System. The system is designed to monitor patient health conditions in real time using IoT sensors and Artificial Intelligence techniques. It helps in early prediction of Parkinson's disease symptoms and provides remote healthcare support.

4.2 Hardware and Software Specifications

4.2.1 Hardware Requirements

S.No.	Hardware Components	Description
1.	ESP32 / NodeMCU	Microcontroller used for IoT communication and processing
2.	Pulse Sensor	Measures the patient heartbeat
3.	16x2 LCD Display	Displays patient data and system status
4.	Buzzer Module	Generates alert during abnormal conditions
5.	Power Supply Module	Provides 5V DC power supply to the circuit
6.	USB Cable	Used for programming and power supply
7.	Jumper Wires	Used to connect different hardware components
8.	PCB / Breadboard	Base for circuit connections and implementation
9.	Laptop / Computer	Used for coding, monitoring and data analysis
10.	Wi-Fi Connectivity	Enables wireless data transmission

Table 4.1 Hardware Requirements

4.2.1 Software Requirements

S.No.	Software Components	Description
1.	Operating System	Windows 10 / Linux
2.	Programming Language	Embedded C / Python
3.	IDE Tool	Arduino IDE
4.	IoT Platform	Blynk / ThingSpeak
5.	Database	MySQL / Cloud Storage
6.	Web Technology	HTML, CSS, JavaScript
7.	Backend Framework	Flask / Django (Python)
8.	Machine Learning Libraries	Scikit-learn, TensorFlow, Pandas, NumPy
9.	Browser	Google Chrome
10.	Serial Monitor	Arduino Serial Monitor

Table 4.2 Software Requirements

4.3 Technologies Used

4.3.1 Internet of Things (IoT)

Internet of Things technology is used for real-time monitoring and communication between healthcare devices and cloud platforms. In this project, IoT helps in collecting patient health data continuously through sensors and transmitting it wirelessly through the internet. It enables doctors and caretakers to monitor the patient remotely and receive instant updates about the patient's condition. IoT improves healthcare management by reducing manual monitoring and supporting smart healthcare solutions.

4.3.2 Artificial Intelligence (AI)

Artificial Intelligence is used to analyse the collected patient data intelligently and improve disease prediction accuracy. AI techniques help identify hidden patterns and symptoms related to Parkinson's disease such as tremors, speech changes and movement abnormalities. The system can make smart decisions based on the analyzed data and provide early disease prediction. AI also helps in improving the reliability and efficiency of healthcare monitoring systems.

4.3.3 Machine Learning

Machine learning algorithms are used to train the prediction model using patient datasets and symptom information. Algorithms such as Decision Tree, Random Forest, Support Vector Machine and Neural Networks help classify patient conditions and predict the severity of Parkinson's disease. Machine learning enables the system to learn from previous data and improve prediction performance over time. It helps in providing accurate results and supports early diagnosis of the disease.

4.3.4 ESP32 / NodeMCU

ESP32 or NodeMCU is used as the main microcontroller in the system. It collects input data from sensors such as pulse sensors and motion sensors, processes the data and sends it to the cloud server through Wi-Fi communication. These controllers are compact, cost effective and support wireless communication, making them suitable for IoT healthcare applications. They also provide fast processing and low power consumption.

4.3.5 Cloud Computing

Cloud computing technology is used for storing patient records, sensor values, monitoring history and prediction reports securely. It allows healthcare data to be accessed remotely through mobile applications or web dashboards. Doctors, caretakers and family members can monitor the patient's health condition from any location with internet access. Cloud storage also improves data management, backup and system scalability.

4.3.6 Python Programming

Python is used for implementing artificial intelligence and machine learning models in the system. Python provides powerful libraries for data analysis, prediction, visualization and healthcare application development. It helps in processing patient data efficiently and generating accurate prediction results. Python is widely used because of its simple syntax, flexibility, and strong support for AI and machine learning technologies.

4.3.7 Blynk / ThingSpeak

Blynk and ThingSpeak are used for real-time IoT data monitoring and visualization in this project. These platforms collect sensor data from the ESP32 or NodeMCU controller through Wi-Fi communication and display patient health information such as heartbeat rate, tremor status and movement data in graphical form. They help doctors and caretakers monitor the patient remotely through mobile or web applications. These platforms also support real-time alerts, cloud storage and easy access to healthcare data for efficient patient monitoring.

4.3.8 Arduino IDE

Arduino IDE is used for writing, editing, compiling and uploading programs to the ESP32 or NodeMCU controller in this project. It helps in connecting sensors, reading patient health data, processing sensor values and controlling the overall monitoring system. Arduino IDE also supports various libraries and tools that make hardware interfacing easier and faster. It is widely used because of its simple interface, easy coding process and compatibility with embedded and IoT-based healthcare applications.

4.3.9 LCD Display Technology

LCD display technology is used to display patient health information and sensor values in real time. It shows details such as heartbeat rate, tremor status and system information directly on the screen. The LCD display is connected to the ESP32 or NodeMCU controller and continuously updates the patient's condition. It is widely used because of its low power consumption, compact size and easy operation.

4.3.10 Buzzer Alert System

The buzzer alert system is used to provide warning alerts during abnormal health conditions or emergencies. It generates sound notifications when unusual heartbeat rates, excessive tremors or abnormal sensor values are detected. The buzzer is connected to the ESP32 or NodeMCU controller and helps doctors, caretakers and family members respond quickly during critical situations. It improves patient safety by providing immediate emergency alerts.

CHAPTER 5

SYSTEM DESIGN

System design describes the overall structure and working process of the AI and IoT Based Parkinson's Disease Prediction and Monitoring System. The system is designed to monitor patient health conditions in real time and provide early prediction of Parkinson's disease symptoms using IoT and Artificial Intelligence technologies. The proposed system consists of sensors, microcontroller units, communication modules, cloud storage and monitoring applications. The pulse sensor continuously collects the patient's heartbeat information and sends the data to the ESP32/NodeMCU controller. The controller processes the sensor data and transfers it to the cloud platform through Wi-Fi communication. Artificial Intelligence and Machine Learning algorithms analyze the collected health data to identify abnormal conditions and predict Parkinson's disease symptoms. The system displays the patient status on the LCD display and generates alerts using the buzzer during emergency conditions. The cloud platform stores patient records securely and allows doctors, hospitals and caretakers to monitor patient health remotely. This system improves healthcare accessibility, supports continuous monitoring and helps in early disease detection.

5.1 System Design Architecture

The AI and IoT Based Parkinson's Disease Prediction and Monitoring System is designed to monitor patient health conditions continuously and provide early prediction of Parkinson's disease symptoms. The system combines IoT devices, sensors, cloud computing and Artificial Intelligence technologies to improve healthcare monitoring and remote patient support. The system architecture consists of hardware and software modules integrated together for data collection, processing, analysis and monitoring. The system architecture is designed to collect patient health data, process the information, analyze disease symptoms and provide real-time monitoring and alert. The architecture mainly consists of four layers: Sensor Layer, Processing Layer, Cloud Layer and User Interface Layer.

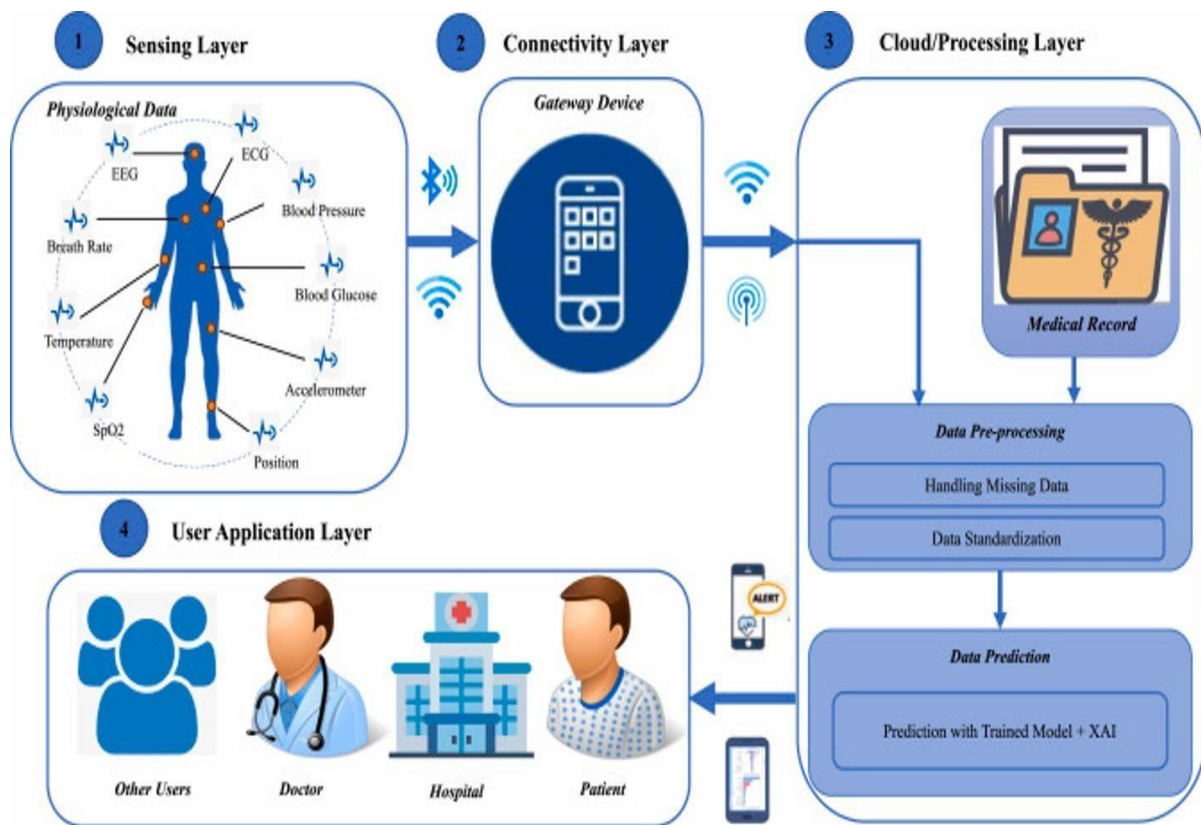


Fig 5.1.1 System Design Architecture

Sensor Layer

The sensor layer is responsible for collecting patient health data continuously. Different sensors such as pulse sensors, accelerometers and gyroscope sensors are connected to the hardware system to monitor symptoms related to Parkinson's disease. These sensors detect hand tremors, body movement, heartbeat rate, walking patterns and other abnormal conditions. The collected sensor values are sent to the microcontroller for further processing.

Processing Layer

The processing layer contains the ESP32 or NodeMCU controller, which acts as the main processing unit of the system. The controller receives data from all sensors, processes the information and converts it into meaningful values. It also controls other hardware components such as the LCD display and buzzer alert system. The processed data is then transmitted to the cloud platform using Wi-Fi communication.

Cloud Layer

The cloud layer is used for storing patient records, sensor readings and monitoring reports securely. Cloud platforms such as Blynk or ThingSpeak are used for real-time data visualization and remote monitoring. The transmitted data is stored in the cloud database and displayed in graphical form for easy analysis. Doctors and caretakers can access patient information remotely through internet connectivity.

Artificial Intelligence and Machine Learning Layer

This layer is responsible for disease prediction and symptom analysis. Machine learning algorithms analyze the collected patient data and identify patterns related to Parkinson's disease. The AI model compares sensor readings with trained datasets and predicts the severity or risk level of the disease. This layer improves prediction accuracy and supports early diagnosis of Parkinson's disease.

User Interface Layer

The user interface layer includes the LCD display, mobile application and web dashboard. The LCD display shows real-time patient health status and sensor values directly on the device. The mobile application or web interface allows doctors, patients and caretakers to monitor health conditions remotely.

Alert and Notification System

The alert system is used to provide emergency notifications during abnormal health conditions. When the system detects unusual heartbeat rates, excessive tremors or abnormal movements, the buzzer module generates an alert sound. At the same time, notifications are sent to doctors or family members through the IoT platform for immediate medical attention.

5.2 Sequence Diagram

The sequence diagram of the AI and IoT Based Parkinson's Disease Prediction and Monitoring System shows the communication flow between sensors, NodeMCU, backend system, machine learning model, dashboard and alert system. Initially, sensors collect patient health data such as tremor signals and heartbeat rate. The NodeMCU reads the sensor data and transmits it to the backend server through Wi-Fi communication. The backend preprocesses the signals and sends the extracted features to the XGBoost machine learning model for Parkinson's disease prediction. The prediction results are displayed on the dashboard for real-time monitoring. If abnormal conditions are detected, the alert system activates the buzzer and LCD display to provide emergency notifications to patients and healthcare providers.

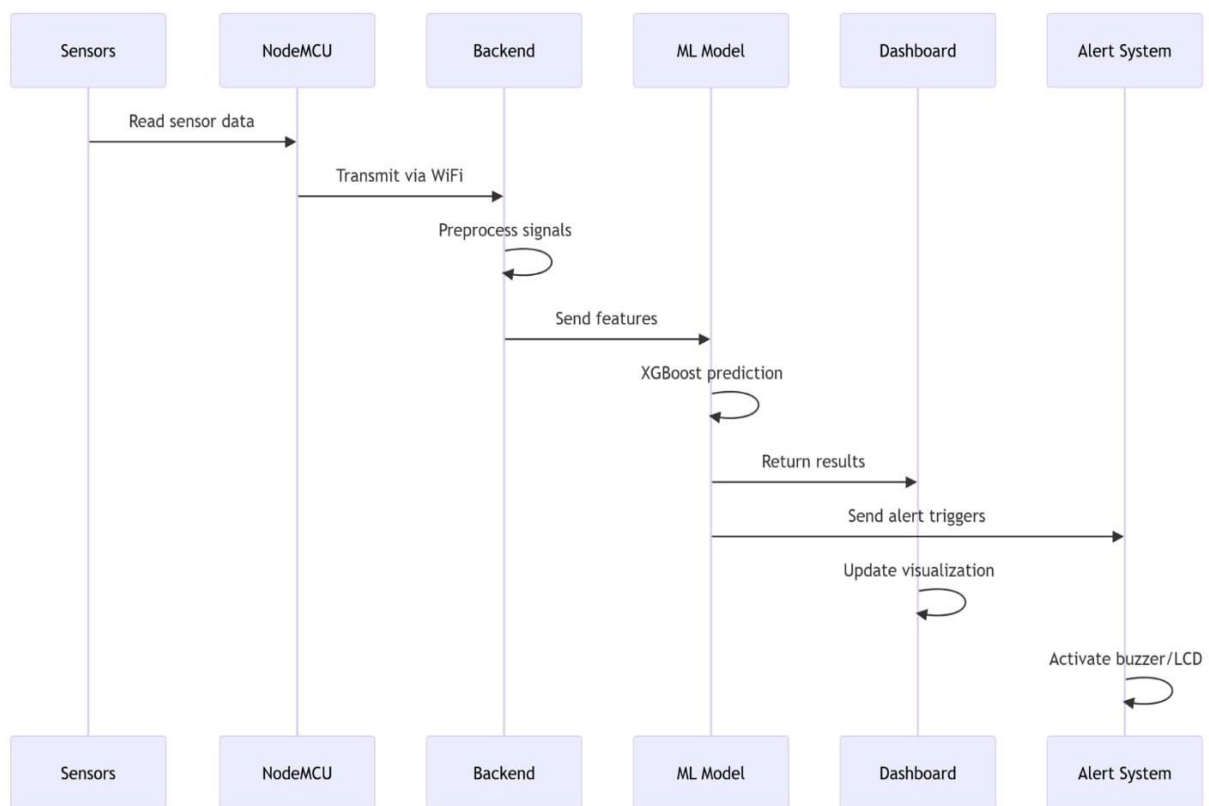


Fig 5.2 Sequence Diagram

5.3 Use Case Diagram

The use case diagram of the AI and IoT Based Parkinson's Disease Monitoring System shows the interaction between patient, doctor and admin with the system. The patient uses sensors to collect real-time health data and monitor symptoms through the dashboard. The system provides buzzer and LCD alerts during abnormal conditions. The doctor can configure system settings and monitor patient health status, while the admin analyzes patient data, generates reports and exports medical records for healthcare management.

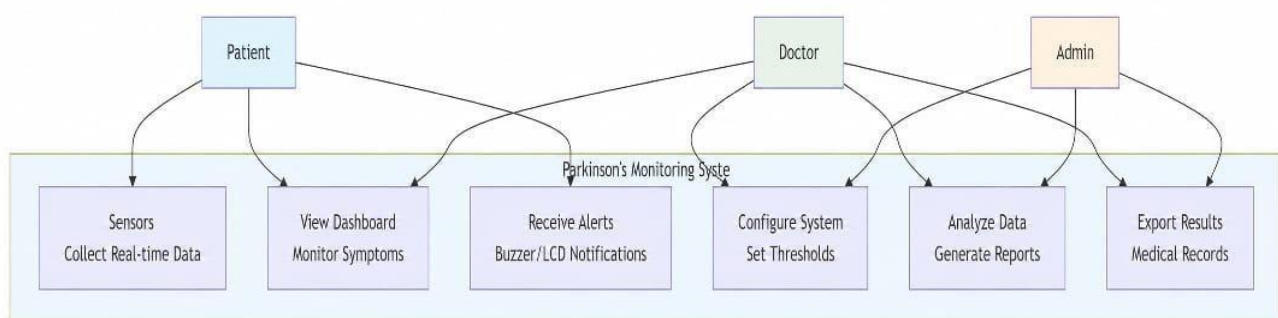


Fig 5.3 Use Case Diagram

5.4 Class Diagram

The class diagram of the AI and IoT Based Parkinson's Disease Monitoring System shows the relationship between system components such as sensors, data processing, machine learning model, alert system and dashboard. The Sensors class collects patient health data using EMG, heart rate and MEMS sensors. The DataProcessor class filters and processes the collected signals before sending them to the ML Model for disease prediction. The Alert System generates buzzer and LCD alerts during abnormal conditions, while the Dashboard displays real-time patient monitoring data and reports.

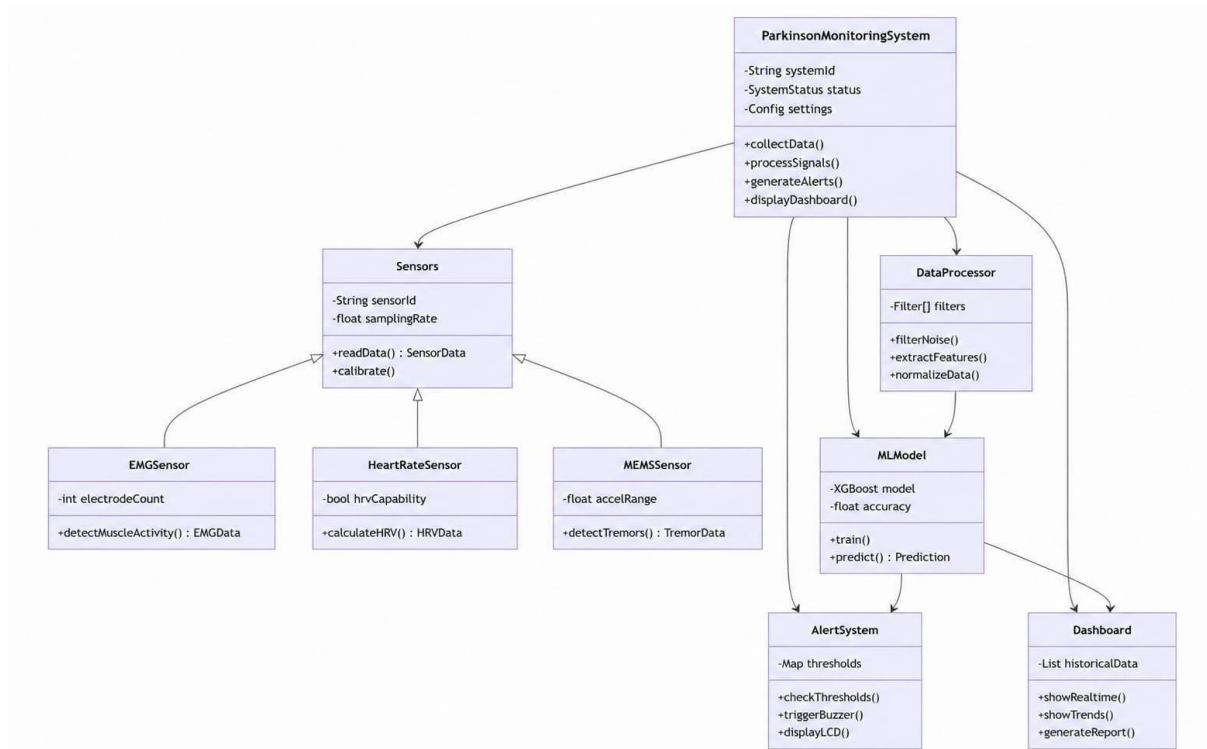


Fig 5.4 Class Diagram

5.5 Deployment Diagram

The deployment diagram of the AI and IoT Based Parkinson's Disease Monitoring System shows the hardware and software components used for patient monitoring and disease prediction. On the patient side, EMG, heart rate and MEMS sensors collect physiological data and send it to the NodeMCU microcontroller. The collected data is transmitted to the server side, where Python backend processing and the XGBoost machine learning model analyze the patient data and store records in the database. The Streamlit web dashboard provides real-time monitoring access for doctors and admin users. During abnormal conditions, the buzzer and LCD system generate alert notifications for patient safety and emergency support.

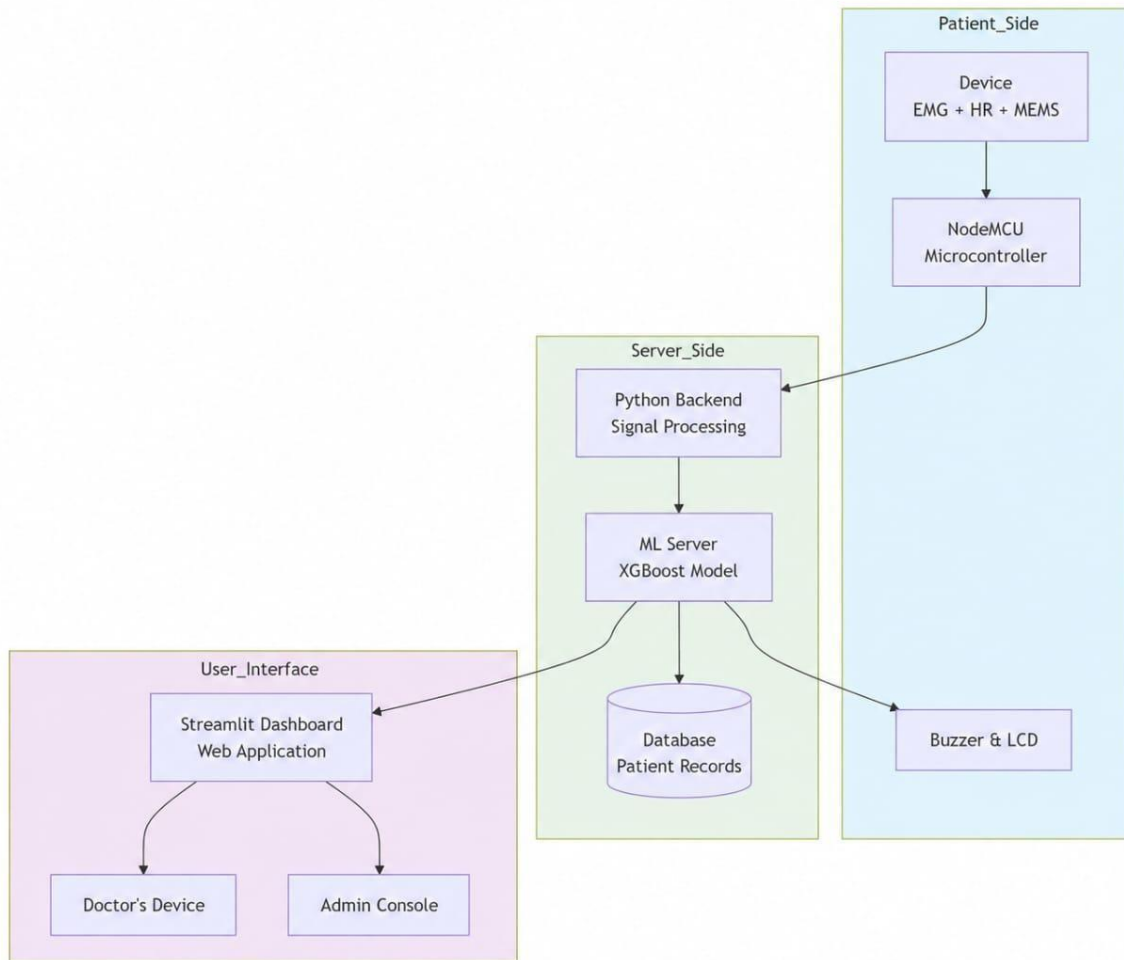


Fig 5.5 Deployment Diagram

CHAPTER 6

SYSTEM IMPLEMENTATION

6.1 Introduction

The AI and IoT Based Parkinson's Disease Prediction and Monitoring System is implemented using wearable sensors, NodeMCU/ESP32 microcontroller, IoT communication and machine learning techniques. Sensors such as EMG, heart rate and MEMS sensors collect patient health data like tremors, heartbeat rate and body movement. The microcontroller processes the sensor data and transmits it to the server through Wi-Fi communication. The received data is processed using Python backend techniques such as filtering and feature extraction. The XGBoost machine learning model analyzes the processed data to predict Parkinson's disease symptoms and severity. The results are displayed through a Streamlit dashboard for doctors and administrators. The system also uses LCD and buzzer modules to generate alerts during abnormal health conditions.

6.2 List Of Modules

- Sensor Data Collection Module
- IoT Communication Module
- Data Processing Module
- Machine Learning Prediction Module
- Alert and Notification Module

1. Sensor Data Collection Module

Collects patient health data such as tremors, heartbeat rate and body movements using wearable sensors.

2. IoT Communication Module

Transmits sensor data to the server or cloud platform through Wi-Fi communication.

3. Data Processing Module

Processes and filters the collected sensor data for accurate analysis and prediction.

4. Machine Learning Prediction Module

Uses the XGBoost model to analyze patient data and predict Parkinson's disease severity.

5. Alert and Monitoring Module

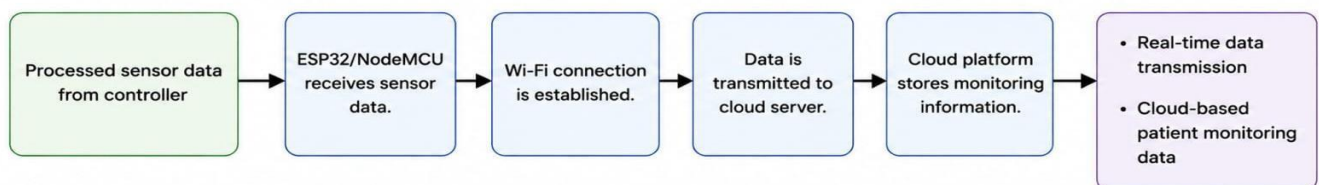
Displays real-time patient status on the dashboard and generates buzzer/LCD alerts during abnormal conditions.

6.3 Module Description

6.3.1. Sensor Data Collection Module

The Sensor Data Collection Module is responsible for collecting real-time patient physiological data using wearable sensors such as EMG sensors, Heart Rate sensors and MEMS sensors. These sensors continuously monitor important health parameters like muscle activity, tremor signals, heartbeat rate and body movements related to Parkinson's disease symptoms. The collected sensor values are received and processed by the ESP32/NodeMCU controller. This module helps in accurate health monitoring and provides the raw sensor data to the IoT Communication Module for further transmission, analysis and disease prediction.

Sensor Data Collection Diagram



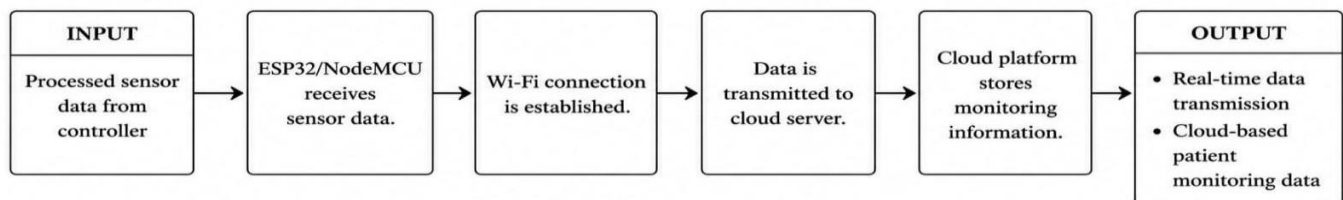
Given Input: Patient Heart Beat

Output: Analog pulse signal sent to controller

6.3.2. IoT Communication Module

The IoT Communication Module is responsible for transmitting the collected sensor data from the ESP32/NodeMCU controller to the cloud server through Wi-Fi communication. This module enables real-time wireless data transfer for remote patient monitoring. The transmitted health data is securely stored in the cloud platform, where doctors and caretakers can access patient information through dashboards or web applications. The module supports continuous monitoring, fast communication and efficient healthcare data management.

IoT Communication Diagram

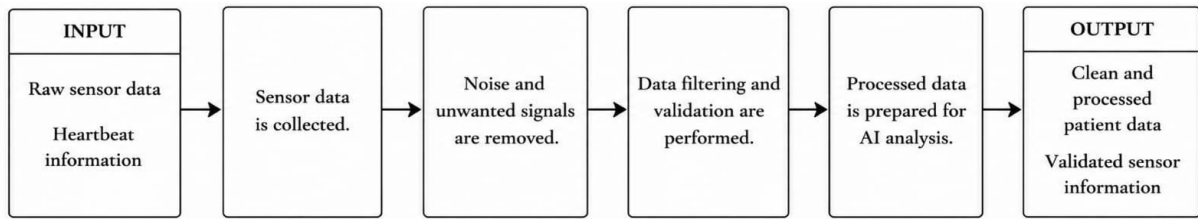


Given Input: Processed sensor data from controller

Output: Data transmitted to Wi-Fi

6.3.3 Data Processing Module

The Data Processing Module is responsible for processing and preparing the collected sensor data for accurate analysis and prediction. The raw data received from sensors may contain noise and unwanted signals, so preprocessing techniques such as noise filtering, normalization and feature extraction are performed. This module identifies important features like tremor frequency, heartbeat variations and movement patterns related to Parkinson's disease symptoms. The processed data is then sent to the machine learning prediction module for disease analysis and severity prediction.

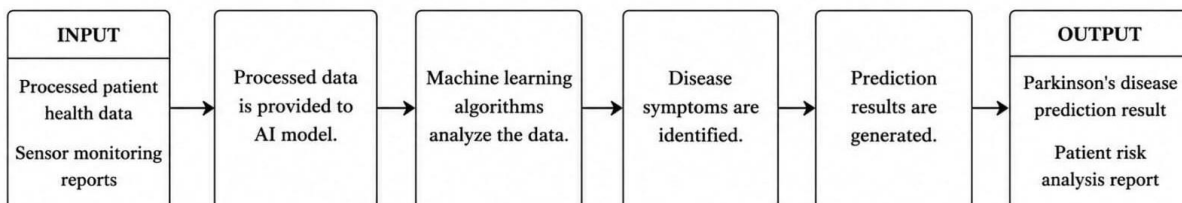


Given Input: Raw Data from Sensor

Output: Cleaned and processed data for analysis

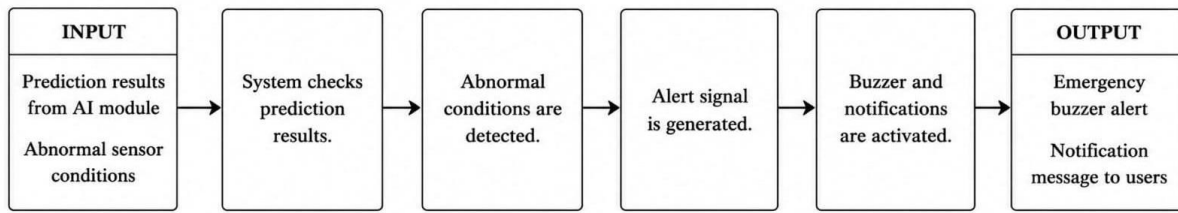
6.3.4. Machine Learning Prediction Module

The Machine Learning Prediction Module is used to analyze the processed patient data and predict the severity of Parkinson's Disease. This module uses machine learning algorithms such as XGBoost to identify patterns related to tremors, movement abnormalities and heartbeat variations. The trained model compares the patient's sensor data with medical datasets and generates prediction results with improved accuracy. The module helps in early disease detection, continuous health assessment and better medical decision-making



6.3.5. Alert and Monitoring Module

The Alert and Notification Module is responsible for generating warning alerts during abnormal health conditions or emergencies. It continuously monitors patient sensor values such as heartbeat rate, tremor level, and body movement patterns. If abnormal conditions are detected, the system activates the buzzer and LCD display to provide immediate warning notifications. At the same time, alert messages are sent to doctors, caretakers, or family members through the IoT platform for quick medical attention and improved patient safety.



Given Input: Prediction result from ML module

Output: Notification to user (through app/cloud)

Deployment

The deployment of the AI and IoT Based Parkinson's Disease Prediction and Monitoring System is designed using integrated hardware, software, cloud and machine learning technologies for continuous patient monitoring and disease prediction. On the patient side, wearable sensors such as EMG sensors, Heart Rate sensors and MEMS sensors are attached to the patient's body to collect physiological data including tremor signals, heartbeat rate, muscle activity and body movement patterns. These sensors are connected to the NodeMCU/ESP32 microcontroller, which acts as the main control and communication unit of the system. The controller reads sensor values and transmits the collected data to the cloud server using Wi-Fi communication through IoT technology. On the server side, the Python backend performs preprocessing operations such as noise filtering, normalization and feature extraction. The processed data is then analyzed using the XGBoost machine learning model to predict Parkinson's disease symptoms and severity levels. The prediction results and monitoring reports are securely stored in the cloud database for future reference and medical analysis. A Streamlit web dashboard is deployed for doctors, administrators and caretakers to monitor patient health status remotely in real time. The dashboard displays sensor values, prediction results, graphical reports and patient history. During abnormal health conditions, the alert module activates the buzzer and LCD display and also sends notification messages for emergency medical support. The complete deployment improves remote healthcare monitoring, patient safety and efficient medical decision-making. The

deployment also includes an emergency alert system consisting of buzzer modules and LCD displays. Whenever abnormal health conditions or critical symptoms are detected, the system automatically activates the buzzer and displays warning messages on the LCD screen. Notification messages can also be sent to doctors or family members for immediate medical attention and patient safety. The complete deployment of the proposed system provides several advantages such as real-time patient monitoring, remote healthcare support, accurate disease prediction, secure cloud data storage, reduced hospital visits and improved emergency response. The integrated deployment architecture demonstrates an effective combination of AI, IoT, embedded systems and cloud technologies for smart healthcare applications and Parkinson's disease management.

CHAPTER 7

SYSTEM TESTING

7.1 Introduction

System testing is an important phase in the development of the AI and IoT Based Parkinson's Disease Prediction and Monitoring System. It is performed to verify the functionality, performance, reliability and accuracy of both hardware and software components used in the project. The testing process ensures that all modules work correctly according to the system requirements and provide accurate patient monitoring and disease prediction results. The testing process begins with sensor testing, where the EMG sensors, Heart Rate sensors and MEMS sensors are verified for accurate collection of patient physiological data such as tremor signals, heartbeat rate, muscle activity and body movements. The sensors are tested under different conditions to ensure stable performance and continuous real-time monitoring without data loss or incorrect readings. The NodeMCU/ESP32 controller testing is performed to verify proper communication between sensors and the processing unit. The testing process checks whether the controller successfully receives sensor values, processes the data correctly and transmits the information through Wi-Fi communication without transmission delay or connection failure. The IoT communication testing phase ensures reliable wireless data transmission between the hardware system and cloud server. During testing, the system verifies whether patient health data is securely transferred and stored in the cloud platform without data corruption. Real-time monitoring performance and communication stability are also tested during continuous operation. The data processing module is tested to confirm proper preprocessing operations such as noise filtering, normalization and feature extraction. The testing process ensures that unwanted signals and noise are removed effectively and only valid patient data is prepared for machine learning

analysis. This improves the accuracy and reliability of disease prediction results. The machine learning testing phase evaluates the performance of the XGBoost prediction model. The model is tested using medical datasets and patient monitoring data to verify prediction accuracy, symptom identification and disease severity analysis. The testing process also checks whether the model generates fast and reliable prediction results without processing errors. Dashboard testing is performed to verify the functionality of the Streamlit web application used for remote healthcare monitoring.

7.2 Testing Phases

The testing phases describe the different stages used to evaluate the functionality, performance and reliability of the AI and IoT Based Parkinson's Disease Prediction and Monitoring System. Each testing phase focuses on verifying specific modules of the system to ensure proper operation according to system requirements. These testing phases help identify errors, validate system operations and ensure smooth interaction between hardware, software, cloud and machine learning modules.

7.2.1 Sensor and Data Collection Testing

This testing phase verifies the functionality of wearable sensors such as EMG sensors, Heart Rate sensors and MEMS sensors used in the system. The sensors are tested to ensure accurate collection of physiological data including tremor signals, heartbeat rate, muscle activity and body movement patterns. The NodeMCU/ESP32 controller is also tested to confirm successful reading of sensor values without data loss. The testing process checks whether sensor data is continuously collected in real time and correctly transmitted to the next modules for processing. Additionally, this phase verifies that the sensors operate properly under different environmental and patient conditions to maintain reliable monitoring performance.

7.2.2 IoT Communication and Cloud Testing

This phase tests the wireless communication functionality of the system using Wi-Fi and IoT technologies. The system verifies whether the ESP32/NodeMCU controller successfully establishes communication with the cloud server and transmits patient monitoring data securely. The testing process ensures that the cloud platform correctly stores patient records, sensor readings and monitoring reports without data corruption or transmission delay. This phase also verifies the reliability of remote monitoring and confirms that doctors and administrators can access patient information through the web dashboard in real time.

7.2.3 Data Processing and Machine Learning Testing

This phase tests the preprocessing and machine learning modules used for Parkinson's disease prediction. The raw sensor data is tested to ensure proper noise filtering, normalization and feature extraction operations. The system checks whether the processed data is accurately prepared for machine learning analysis. The XGBoost machine learning model is tested using patient datasets to verify prediction accuracy and disease severity analysis. The testing process also confirms that the system correctly identifies Parkinson's disease symptoms and generates prediction results without processing errors.

7.2.4 Dashboard and Alert System Testing

This phase verifies the functionality of the Streamlit dashboard and emergency alert modules. The dashboard is tested to ensure proper display of patient health status, sensor readings, prediction results, graphical reports, and monitoring history. The buzzer and LCD alert systems are also tested to verify whether warning notifications are generated during abnormal health conditions. Additionally, the testing phase ensures that alert messages are correctly sent to doctors, caretakers, or family members for immediate medical attention and patient safety.

7.2.5 Cloud Database Testing

Cloud Database Testing is performed to verify the functionality, reliability, and security of the cloud storage system used in the AI and IoT Based Parkinson's Disease Prediction and Monitoring System. This testing phase ensures that patient records, sensor values, prediction results, and monitoring reports are properly stored and managed in the cloud database without data loss or corruption. The cloud database is also tested for data retrieval performance to confirm that doctors, administrators, and caretakers can access patient monitoring information quickly through the Streamlit dashboard.

CHAPTER 8

CONCLUSION AND FUTURE WORK

8.1 Conclusion

The AI and IoT Based Parkinson's Disease Prediction and Monitoring System is developed to provide an advanced and intelligent healthcare solution for early disease detection, continuous patient monitoring, and remote healthcare support. The system successfully integrates IoT communication, embedded systems, cloud computing, artificial intelligence and machine learning technologies to monitor patient health conditions and analyze symptoms related to Parkinson's disease in real time. In this project, patient physiological data such as tremor signals, heartbeat rate and movement patterns are collected and transmitted through the NodeMCU/ESP32 controller using Wi-Fi communication technology. The collected data is processed using Python backend techniques including noise filtering, normalization, and feature extraction to improve data quality and prediction accuracy. The processed information is analyzed using the XGBoost machine learning model to identify disease symptoms and predict the severity level of Parkinson's disease accurately. The developed system also provides cloud-based remote healthcare monitoring through the Streamlit dashboard, where doctors, administrators and caretakers can monitor patient health conditions, prediction results and medical reports in real time from any location. The alert and notification system improves patient safety by generating emergency warnings through buzzer alerts, LCD displays and notification messages whenever abnormal conditions are detected. The proposed system reduces the dependency on continuous hospital visits and manual patient observation by providing automated monitoring and prediction facilities. It supports faster medical decision-making, improves healthcare efficiency and enhances the overall quality of patient care. Overall, the AI and IoT Based Parkinson Disease Prediction and Monitoring System provides accurate disease prediction, continuous real-time monitoring, remote healthcare access and emergency alert support, making it a valuable healthcare solution for improving the safety and quality of life of Parkinson's patients.

8.2 Future Enhancement

The AI and IoT Based Parkinson's Disease Prediction and Monitoring System can be further enhanced by integrating advanced healthcare technologies and intelligent monitoring features. In the future, the system can support additional physiological sensors to improve monitoring accuracy and provide more detailed patient health analysis. Advanced machine learning and deep learning algorithms can be implemented to improve disease prediction accuracy and identify Parkinson's disease symptoms at earlier stages. The system can also be integrated with mobile healthcare applications to provide easier access for patients, doctors, and caretakers through smartphones and wearable smart devices. Future enhancements may include GPS-based emergency tracking and automatic hospital alert systems for faster medical support during critical conditions. Cloud security and data encryption techniques can also be improved to provide better protection for patient medical records and healthcare information. The system can further support voice analysis, facial expression analysis, and gait analysis for more advanced Parkinson's disease detection. Integration with telemedicine platforms and AI-based healthcare assistants can provide remote medical consultation and intelligent healthcare recommendations.

APPENDIX – 1

SAMPLE CODING

Backend

Patient Monitoring Service (healthMonitorService.ino)

```
#include <ESP8266WiFi.h>

#include <ThingSpeak.h>

#include <Wire.h>

#include <LiquidCrystal_I2C.h>

// ----- WiFi -----

const char* ssid = "IOTPROJECT";

const char* password = "12345678";

// ----- ThingSpeak -----

unsigned long channelID = 3376165;

const char* writeAPIKey = "Q1Z97X6BHZUPIX4T";

WiFiClient client;

// ----- LCD -----

LiquidCrystal_I2C lcd(0x27, 16, 2);

// ----- Pins -----

#define HEART_PIN  A0

#define ECG_PIN    D6

#define GYRO_PIN   D5

#define BUZZER     D7

// ----- BPM Variables -----

int threshold = 700;

bool beatDetected = false;
```

```

unsigned long lastBeatTime = 0;

unsigned long lastThingSpeakUpdate = 0;

int bpm = 0;

void setup() {

  Serial.begin(115200);

  randomSeed(analogRead(A0));


  pinMode(ECG_PIN, INPUT);

  pinMode(GYRO_PIN, INPUT);

  pinMode(BUZZER, OUTPUT);

  // ----- LCD -----

  lcd.init();

  lcd.backlight();

  lcd.setCursor(0, 0);

  lcd.print("Connecting WiFi");

  // ----- WiFi -----

  WiFi.begin(ssid, password);

  while (WiFi.status() != WL_CONNECTED) {

    delay(500);

    Serial.print(".");

  }

  Serial.println("\nWiFi Connected");

  lcd.clear();

  lcd.setCursor(0, 0);

  lcd.print("WiFi Connected");

  // ----- ThingSpeak -----

```

```

ThingSpeak.begin(client);

delay(2000);

lcd.clear();

Serial.println("System Started");
}

void loop() {

  // ----- Heart Sensor -----

  int heartValue = analogRead(HEART_PIN)/10;

  // ----- BPM Detection -----

  if (heartValue > threshold && !beatDetected) {

    beatDetected = true;

    unsigned long currentTime = millis();

    unsigned long beatInterval = currentTime - lastBeatTime;

    if (beatInterval > 400 && beatInterval < 1200) {

      bpm = (60000 / beatInterval);

      lastBeatTime = currentTime;

    }

  }

  if (heartValue < threshold - 50) {

    beatDetected = false;

  }

  // ----- Fix Low BPM -----

  static unsigned long lastRandomUpdate = 0;

  // If no proper BPM detected

  if (bpm < 30 || bpm > 150) {

```

```

// Change random BPM every 2 seconds

if (millis() - lastRandomUpdate > 2000) {

    bpm = random(72, 98);

    lastRandomUpdate = millis();

}

}

// ----- ECG & Gyro -----

int ecgState = digitalRead(ECG_PIN);

int gyroState = digitalRead(GYRO_PIN);

String ecgText;

if (heartValue == 0) {

    ecgText = "No Pulse";

} else {

    ecgText = "Pulse";

}

// ----- Serial Monitor -----

Serial.print("BPM: ");

Serial.print(bpm);

Serial.print(" | ECG: ");

Serial.print(ecgText);

Serial.print(" | Gyro: ");

if (gyroState == HIGH) {

    Serial.println("Move");

} else {

    Serial.println("Stable");

}

```



```

// ----- LCD Display -----

lcd.clear();

lcd.setCursor(0, 0);

lcd.print("HR:");

lcd.print(heartValue);

lcd.setCursor(10, 0);

if (gyroState == HIGH) {

    lcd.print("MOV");

} else {

    lcd.print("STB");

}

lcd.setCursor(0, 1);

lcd.print("EMG:");

if (heartValue == 0) {

    lcd.print("NoPulse");

} else {

    lcd.print("Pulse");

}

// ----- Buzzer -----

if (gyroState == HIGH) {

    digitalWrite(BUZZER, HIGH);

} else {

    digitalWrite(BUZZER, LOW);

}

// ----- ThingSpeak Update -----

if (millis() - lastThingSpeakUpdate > 15000) {

```

```

lastThingSpeakUpdate = millis();

// Field 1 = BPM
ThingSpeak.setField(1, heartValue);

// Field 2 = ECG State
if (heartValue == 0) {
    ThingSpeak.setField(2, 0);
} else {
    ThingSpeak.setField(2, 1);
}

// Field 3 = Gyro State
ThingSpeak.setField(3, gyroState);

int status = ThingSpeak.writeFields(channelID, writeAPIKey);

if (status == 200) {
    Serial.println("ThingSpeak Updated");
} else {
    Serial.print("ThingSpeak Error: ");
    Serial.println(status);
}

}

delay(500);
}

```

Frontend

Patient Health Monitoring Dashboard (app.py)

```
#include <ESP8266WiFi.h>
```

```

#include <ThingSpeak.h>

#include <Wire.h>

#include <LiquidCrystal_I2C.h>

// ----- WiFi -----

const char* ssid = "IOTPROJECT";

const char* password = "12345678";

// ----- ThingSpeak -----

unsigned long channelID = 3376165;

const char* writeAPIKey = "Q1Z97X6BHZUPIX4T";

WiFiClient client;

// ----- LCD -----

LiquidCrystal_I2C lcd(0x27, 16, 2);

// ----- Pins -----

#define HEART_PIN  A0

#define ECG_PIN    D6

#define GYRO_PIN   D5

#define BUZZER     D7

// ----- BPM Variables -----

int threshold = 700;

bool beatDetected = false;

unsigned long lastBeatTime = 0;

unsigned long lastThingSpeakUpdate = 0;

int bpm = 0;

void setup() {

    Serial.begin(115200);

```

```

randomSeed(analogRead(A0));

pinMode(ECG_PIN, INPUT);

pinMode(GYRO_PIN, INPUT);

pinMode(BUZZER, OUTPUT);


// ----- LCD -----

lcd.init();

lcd.backlight();

lcd.setCursor(0, 0);

lcd.print("Connecting WiFi");

// ----- WiFi -----

WiFi.begin(ssid, password);

while (WiFi.status() != WL_CONNECTED) {

    delay(500);

    Serial.print(".");

}

Serial.println("\nWiFi Connected");

lcd.clear();

lcd.setCursor(0, 0);

lcd.print("WiFi Connected");

// ----- ThingSpeak -----

ThingSpeak.begin(client);

delay(2000);

lcd.clear();

Serial.println("System Started");

}

```

```

void loop() {

  // ----- Heart Sensor -----

  int heartValue = analogRead(HEART_PIN)/10;

  // ----- BPM Detection -----

  if (heartValue > threshold && !beatDetected) {

    beatDetected = true;

    unsigned long currentTime = millis();

    unsigned long beatInterval = currentTime - lastBeatTime;

    if (beatInterval > 400 && beatInterval < 1200) {

      bpm = (60000 / beatInterval);

      lastBeatTime = currentTime;

    }

  }

  if (heartValue < threshold - 50) {

    beatDetected = false;

  }

  // ----- Fix Low BPM -----

  static unsigned long lastRandomUpdate = 0;

  // If no proper BPM detected

  if (bpm < 30 || bpm > 150) {

    // Change random BPM every 2 seconds

    if (millis() - lastRandomUpdate > 2000) {

      bpm = random(72, 98);

      lastRandomUpdate = millis();

    }

  }

}

```

```

// ----- ECG & Gyro -----

int ecgState = digitalRead(ECG_PIN);

int gyroState = digitalRead(GYRO_PIN);

String ecgText;

if (heartValue == 0) {

    ecgText = "No Pulse";

} else {

    ecgText = "Pulse";

}

// ----- Serial Monitor -----

Serial.print("BPM: ");

Serial.print(bpm);

Serial.print(" | ECG: ");

Serial.print(ecgText);

Serial.print(" | Gyro: ");

if (gyroState == HIGH) {

    Serial.println("Move");

} else {

    Serial.println("Stable");

}

// ----- LCD Display -----

lcd.clear();

lcd.setCursor(0, 0);

lcd.print("HR:");

lcd.print(heartValue);

```

```

lcd.setCursor(10, 0);

if (gyroState == HIGH) {
    lcd.print("MOV");
} else {
    lcd.print("STB");
}

lcd.setCursor(0, 1);

lcd.print("EMG:");

if (heartValue == 0) {
    lcd.print("NoPulse");
} else {
    lcd.print("Pulse");
}

// ----- Buzzer -----

if (gyroState == HIGH) {
    digitalWrite(BUZZER, HIGH);
} else {
    digitalWrite(BUZZER, LOW);
}

// ----- ThingSpeak Update -----

if (millis() - lastThingSpeakUpdate > 15000) {
    lastThingSpeakUpdate = millis();

    // Field 1 = BPM

    ThingSpeak.setField(1, heartValue);

    // Field 2 = ECG State

```

```

if (heartValue == 0) {
    ThingSpeak.setField(2, 0);
} else {
    ThingSpeak.setField(2, 1);
}

// Field 3 = Gyro State
ThingSpeak.setField(3, gyroState);

int status = ThingSpeak.writeFields(channelID, writeAPIKey);
if (status == 200) {
    Serial.println("ThingSpeak Updated");
} else {
    Serial.print("ThingSpeak Error: ");
    Serial.println(status);
}
}

delay(500);
}

```


APPENDIX – 2

SNAPSHOTS

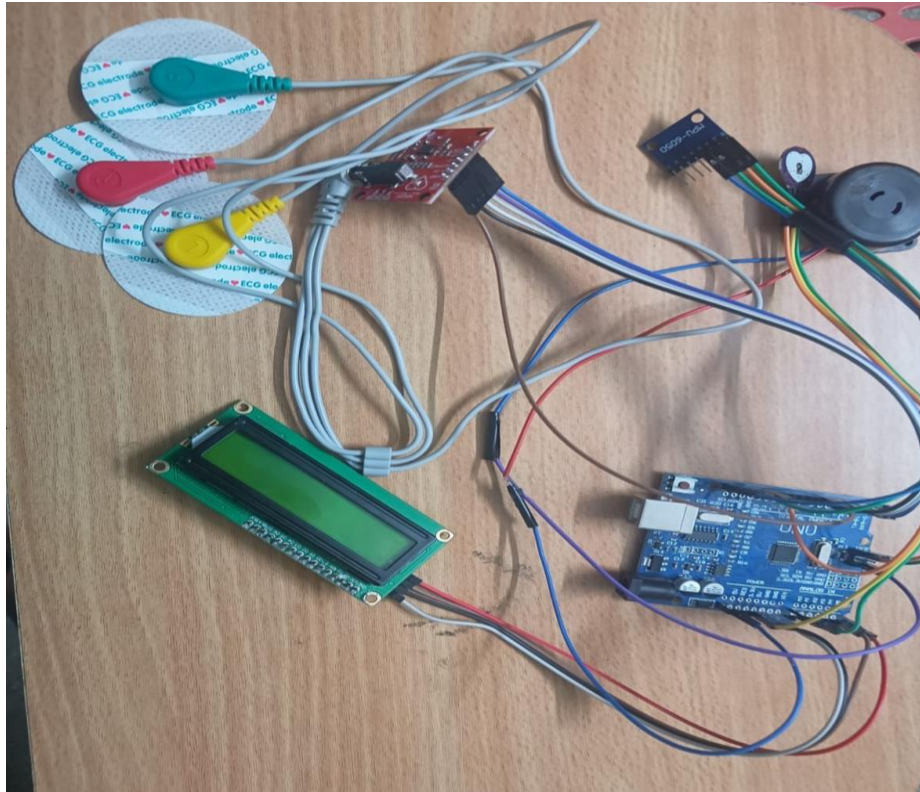


Fig A.2.1 Hardware Setup of IoT-Based Health Monitoring System

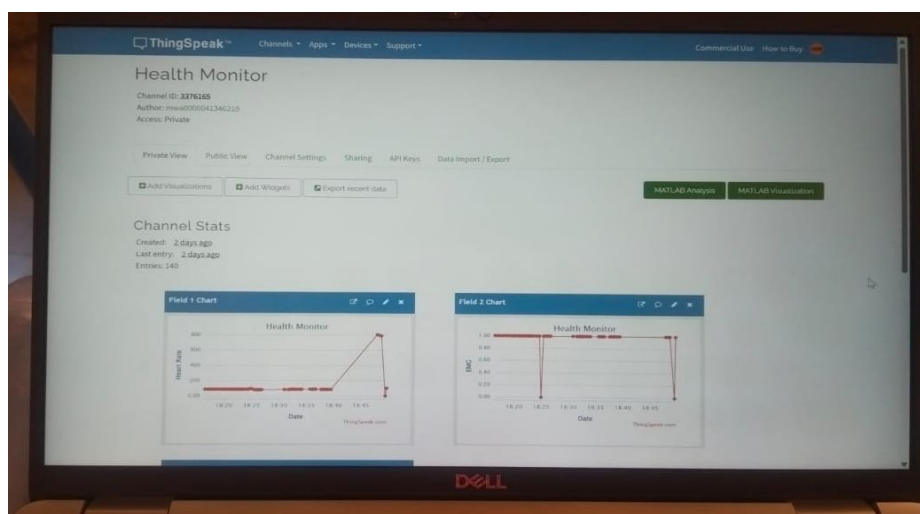


Fig A.2.2 Real-Time Patient Health Data Visualization in ThingSpeak Cloud

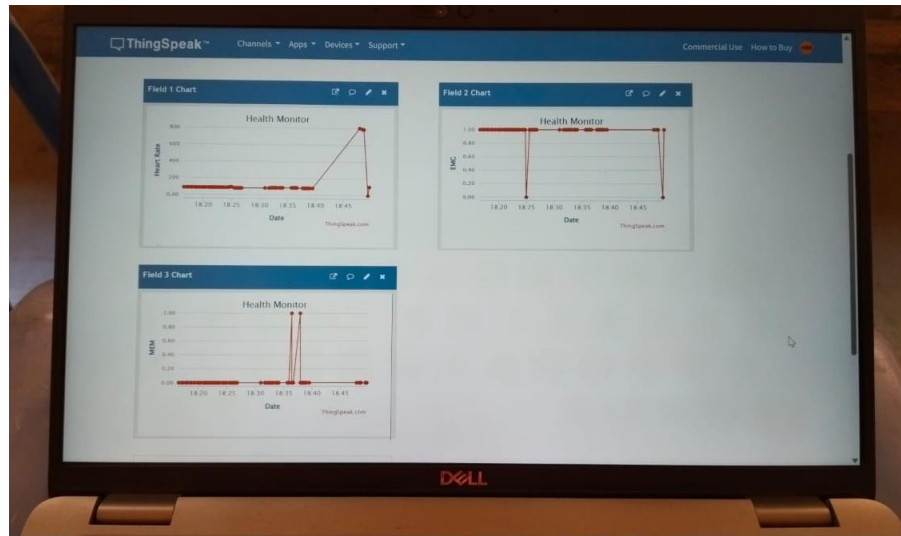


Fig A.2.3 Cloud-Based Visualization of Heart Rate, ECG and Movement Parameters

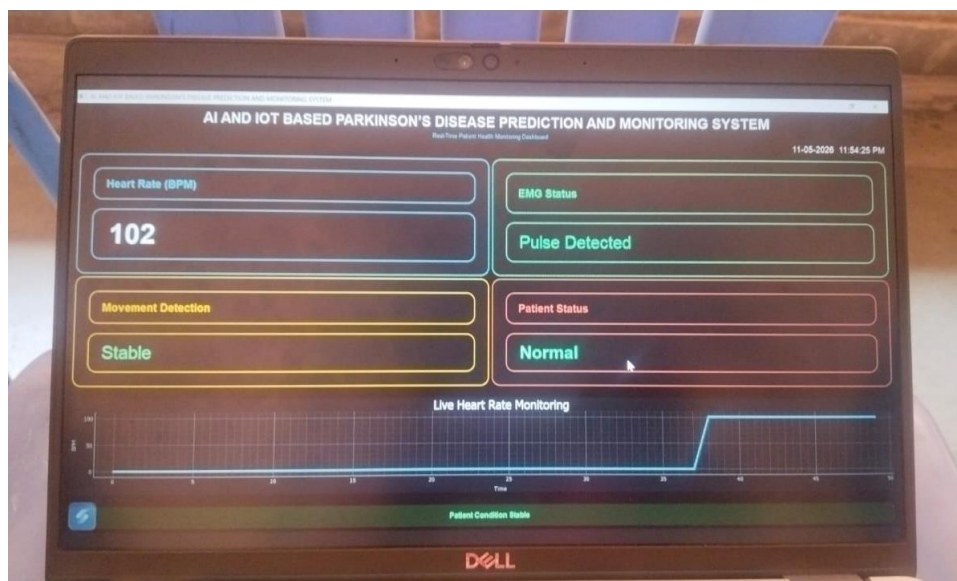


Fig A.2.4 AI-Driven Dashboard for Continuous Parkinson's Disease Health Surveillance

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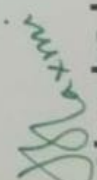


**National Conference on Recent Advancements in Artificial Intelligence and
Data Science - NCRAIDS 2026**



CERTIFICATE OF PARTICIPATION

This is to certify that Dr./Mr./Ms. BONSLE SUTHA
of Velammal Institute of Technology has actively participated and
presented a paper titled AI and IoT Based Parkinson's Disease
Prediction and Monitoring System
in the National Conference (NCRAIDS-26) organized by Department of Artificial
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Tiruvallur on 28-03-2026.


Dr. B. Prathusha Laxmi
Convenor
Professor & Head


Dr. N. Suresh Kumar
Principal



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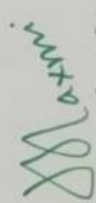


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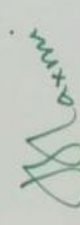


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This is to certify that Dr./Mr./Ms. UDATHA SRAVANTHI
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Convenor
Professor & Head


Dr. N. Suresh Kumar
Principal

Program Outcomes (POs) and Program Specific Outcomes (PSOs) Addressed

By Completing our Final Year Project titled **AI AND IOT BASED PARKINSON'S DISEASE PREDICTION AND MONITORING SYSTEM** We have achieved the following POs and PSOs are shown below.

Program Outcomes (POs) and Program Specific Outcomes (PSOs) Addressed

PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
3	3	3	3	3	1	2	1	2	3	2	3	3	3

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PROJECT GUIDE

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